

CS418 HW10

Title: Homework 10

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9.11A:

Let T be EMST of P , and let pq be an edge of T .

Need to show that the pq is a Delaunay edge.

This will apply for every edge of T , so all EMST edges lie in Delaunay triangulation.

So need to remove EMST edge, look at the cut, use the lightest edge across the cut, then show diameter disk is empty, so edge is Delaunay

1. First need to remove the edge pq from T
 - This will split T into two parts, A and B , and $p \in A$ and $q \in B$
2. pq is the shortest edge crossing the cut (A, B)
 - If an edge ab existed where $a \in A$ and $b \in B$ and $|ab| < |pq|$, then replacing pq by ab would then provide a spanning tree with smaller length.
 - This would contradict that it is an EMST
3. Then consider that a closed disk D has pq as its diameter
4. D then contains no point of P in its interior
 - If Some point $r \in P$ lies inside D , then by Thales theorem $\angle prq > 90^\circ$
 - Therefore the side that is opposite to that angle is the longest, so:

$$|pq| > |pr| \text{ and also } |pq| > |rq|$$

5. Point r lies in A or B
 - If $r \in A$, then edge rq crosses the (A, B) cut and $|rq| < |pq|$. This contradicts the claim of the second step of the proof
 - If $r \in B$, then edge pr crosses the cut (A, B) , and $|pr| < |pq|$. This is also a contradiction of step 2.
6. Therefore the disk with the diameter as pq is empty of all other points P
7. Delaunay edge is exactly an edge $p_i p_j$ for which there exists a closed disk has p_i, p_j on its boundary and no other site is inside
 - So pq is an edge of the Delaunay graph
8. Since every EMST edge is a Delaunay edge, the set of edges of Delaunay triangulation contains an EMST for P

8.1:

Let $p = (p_x, p_y)$

Let the non-vertical line be $\ell : y = mx + b$

Duality transform is defined as $p^* : y = p_x * x - p_y$ and $\ell^* = (m, -b)$

Observation 8.3 states the transform is incidence and order preserving. Need to prove both.

Incidence Preserving:

Need to show that $p \in \ell \iff \ell^* \in p^*$

Now $p \in \ell \iff p_y = mp_x + b$

$\ell^* = (m, -b)$ lies on the line p^* iff the coordinates satisfy the equation of p^*

So, $-b = p_x m - p_y$ and rearranged to $p_y = mp_x + b$ which is what we found above.

So, $\ell^* \in p^* \iff p_y = mp_x + b$

And therefore, $p \in \ell \iff \ell^* \in p^*$

So transform is incidence preserving.

Order Preserving:

Need to show that p lies above $\ell \iff \ell^*$ lies above p^*

p lies above ℓ when $p_y > mp_x + b$

So $\ell^* = (m, -b)$

The point ℓ^* lies above the line p^* iff y coordinate is greater than the value of p^* at $x = m$

So, $-b > p_x m - p_y$

This is rearranged to $p_y > mp_x + b$

Therefore, p above $\ell \iff \ell^*$ above p^*

So the transform is order preserving.

Conclude:

The duality transform

$$(p_x, p_y) \mapsto p^* : y = p_x x - p_y, \quad (y = mx + b) \mapsto \ell^* = (m, -b)$$

Is both incidence and order preserving, and the 8.3 claim is proven true.

8.7:

First do vertical separators.

Vertical line $x = c$ separates R and B iff either $(r \in R) \max r_x < (b \in B) \min b_x$ or $(b \in B) \max b_x < (r \in R) \min r_x$

These four extreme values can be found with one scan and therefore checked in $O(n)$ time.

Now need to look for a non-vertical separator. Do two tests for symmetry:

- All red points above the line and all blue points below it

- All blue points above the line and all red points below it.

If either of above tests is true, then separator exists.

For one of the two tests, suppose we want a line $\ell : y = mx + c$ so that all red points lie above ℓ and all blue points lie below ℓ .

Use duality transform

$$p = (p_x, p_y) \mapsto p^* : y = p_x x - p_y,$$

$$\ell : y = mx + c \mapsto \ell^* = (m, -c)$$

This transform is incidence and order preserving in that p above $\ell \leftrightarrow \ell^*$ above p^*

So the condition that all red points are above ℓ and all blue points are below ℓ in the dual plane is just:

- For all red points $r \in R$, dual point ℓ^* must lie above the line r^*
- For all blue points $b \in B$, dual point ℓ^* must lie below the line b^*

Therefore there are $2n$ half plane constraints in the dual plane:

$$\ell^* \in (r \in R) \bigcap (\text{points above } r^*) \bigcap (b \in B) \bigcap (\text{points below } b^*)$$

Separator of this orientation exists iff the intersection of half planes is nonempty.

Now this is a 2D linear programming feasible problem that has $2n$ constraints.

Through randomized LP algorithm, 2D linear program with N constraints can be solved in $O(N)$ expected time, hence in $O(n)$ expected time.

Algorithm:

1. Check for vertical separator
2. Create the dual half plane system for red above and blue below
3. Run the randomized 2D linear programming feasibility on it
4. Build the dual half plane system for blue points above and red points below
5. Run the randomized 2D linear programming feasibility on it
6. Return yes iff one of the three checks succeeds.

Vertical line test is linear, and since LP problem has $2n$ constraints it is in $O(n)$ expected time. Therefore the algorithm runs in $O(n)$ expected time.

Conclude:

A randomized algorithm exists that decides in $O(n)$ expected time whether R and B have a separator.

8.11:

Let

$$L^+ := \{\ell \in L : \ell \text{ passes above the origin}\} \text{ and } L^- := \{\ell \in L : \ell \text{ passes below origin}\}$$

The origin lies in face f , so no line of L passes through the origin.

Use duality transform that is incidence and order preserving

$$p = (p_x, p_y) \mapsto p^* : y = p_x x - p_y,$$

$$\ell : y = mx + b \mapsto \ell^* = (m, -b)$$

For any point $q \in f$, q lies on the same side of every line $\ell \in L$ as the origin does, since f is one face of the arrangement.

Cases:

- If $\ell \in L^+$, then q is below ℓ
- If $\ell \in L^-$, then q is above ℓ

Through order preservation, these turn into

- If $\ell \in L^+$, then ℓ^* lies below q^*
- If $\ell \in L^-$, then ℓ^* lies above q^*

So, $\{q^* : q \in f\}$ is the set of lines in the dual plane that separate the two dual point sets below:

$$P^+ := \{\ell^* : \ell \in L^+\}$$

$$P^- := \{\ell^* : \ell \in L^-\}$$

All points of P^+ are on one side of the line and all points of P^- are on the other.

Every such dual line has all points of P^+ below it and all points of P^- above it

So duals of the points in f are the separator lines of the two dual point sets

Now let v be vertex of f . Then v is the intersection of two lines $\ell_1, \ell_2 \in L$. Through incidence preservation, the dual v^* is the line through the two dual points ℓ_1^* and ℓ_2^*

Since v is on the boundary of the face that has the origin, v^* is boundary separator for the two dual point sets. There are then three cases for this problem.

1. Both ℓ_1 and ℓ_2 pass below the origin
 - Then $\ell_1, \ell_2 \in L^-$, so $\ell_1^*, \ell_2^* \in P^-$
 - Thus v^* passes through two points of P^- , and all points of P^+ lie on or below v^* . While all points of P^- lie on or above v^* .
 - So v^* would be a lower supporting line of the set P^-
2. ℓ_1 and ℓ_2 pass above the origin
 - Then $\ell_1, \ell_2 \in L^+$, so $\ell_1^*, \ell_2^* \in P^+$
 - Thus v^* passes through two points of P^+ , and all points of P^+ lie on or below v^* , while all points of P^- lie on or above v^*
 - So v^* is upper supporting line of the set P^+ that would still separate P^+ and P^-
3. One of ℓ_1 and ℓ_2 passes above the origin, and the other passes below the origin

- One dual point lies in P^+ and the other lies in P^-
- So v^* passes through one point of each set, with all of P^+ on or below it and all of P^- on or above it
- So v^* is a common supporting tangent separating the two sets.

Therefore if lines of L are split into those passing above and below the origin, then the duals of the points in face f are exactly the lines that separate the two dual point sets.

Also, if v is a vertex of f , then v^* is the separator line through dual points of the two primal lines defining v outlined in the three cases above.

So f dualizes to the set of all separators, and the vertices of f dualize to the extremal separator lines.